Image Tagger
Status Report

Repository state reviewed March 30, 2026

Taggert's summary and recommendation:

Out of what remains to be done, **I would recommend directing the students towards 1. 3D/spatial factors and 2. lighting factors**, because they're critical groups that I don't yet have good methods for. This report notes that there are some gaps we left in room/building type, materials, and affordances while testing, but as I do have good methods for these, I can easily polish them up in the coming weeks with minimal effort. If the students are able to detect all of our spatial and lighting elements by the end of the quarter, we will have a complete prototype with all of the major groups covered.

1. Taxonomy Workbook Scope

The target taxonomy lives in docs/Environment_Cognition_Taxonomy_Hierarchical_V2.7_ActivityAffordances_RoomMarkers copy.xlsx (and also here) It currently contains 16 worksheets spanning high-level building and room categories, activity affordances, latent variables, mediators, environmental attributes, and crosswalk/evidence tables.

The workbook is not a flat tag list — it is a hierarchical research ontology. Key worksheets include:

- Building_Typologies — defines broad program classes
- Room_Typology — defines 45 room/program subtypes with parent building types, typical activities, artifacts, and layout cues
- Activity_Affordances — defines 130 latent activity affordances (L059–L188) with room-type priors and key 2D cues
- Latent_Variables and Mediators — define broader psychological and contextual constructs
- Env_Attributes — catalogs 286 measurable environmental variables with extractability notes, evidence strength, and suggested outcomes/tests

Several additional tabs act as relationship tables rather than tag inventories (e.g., RoomType_to_Activity, Link_Evidence, Attr_to_Outcomes, and the marker-node tabs).

2. Currently Supported Taxonomy Coverage

Room_Typology

The Room_Typology sheet is a 45-row room/program taxonomy keyed by RoomType_ID. Each row includes a room name, parent building typology, typical activities, signature artifacts, signature layout cues, an estimate of 2D inferability, and notes about VR enrichment.

The current system supports this tab in a compressed but operational form. Rather than predicting all 45 room types directly, the live pipeline runs Places365 scene recognition and maps fine scene labels into the 13 coarse room tags from the taxonomy. Further inference to the specific 45 room types will come once a small helper model is trained:

- bathroom, bedroom, corridor, dining, kitchen, living, office
- lobby, retail, classroom, industrial, outdoor_adjacent, other

Implementation lives in `backend/science/vision/room_detection.py` and is invoked from `backend/science/pipeline.py` during canonical processing. Results are persisted as scalar room attributes, a structured `room_json` artifact, and one canonical `room_type.*` tag when the top coarse confidence is at least 0.20.

Activity_Affordances

The Activity_Affordances sheet contains 130 latent affordance rows (L059–L188). Each row records an activity family, activity name, definition, primary room-type priors, key 2D cues, and tier.

The current system operationally supports five affordances:

- L059 — Sleep (Primary)
- L079 — Cook (Daily)
- L091 — Computer Work (Solo)
- L130 — Casual Conversation / Hangout
- L141 — Yoga / Stretching

The method is implemented in `backend/science/context/affordance.py`. The analyzer extracts a 310-dimensional feature vector from segmentation/object-layout cues and evaluates per-affordance LightGBM regressors on a 1–7 Likert scale. Results are packaged into `affordance_json` and high-scoring entries are promoted to canonical affordance tags.

The affordance infrastructure is real and end-to-end — model loading, feature extraction, scoring, artifact persistence, canonical tag derivation, and Explorer API exposure are all present. The remaining obstacle to wider coverage is time and training data, not architecture.

Observable 2D Cues for L059–L188 (Latent_Variables)

The relevant portion of Latent_Variables (L059–L188) mirrors the activity-affordance layer, using the Observable_Cues_2D column to describe the visible evidence supporting each activity judgment.

The current system supports this layer for the same five affordances listed above. The code does not store cues as a hand-authored symbolic checklist; instead, it operationalizes them as:

- Segmentation-derived object presence and counts
- Object area fractions and pairwise distances
- Room-level aggregates and scene topology features

Cue extractors live in `backend/science/context/affordance.py`, using Mask2Former COCO panoptic segmentation when available and a OneFormer ADE20K-to-COCO fallback otherwise. For the remaining 125 affordance rows, cue descriptions exist in the spreadsheet but have not yet been implemented as predictors.

3. Additional Implemented Detections

The following capabilities are implemented in the current repository but are not explicitly enumerated in the taxonomy workbook.

COCO Object Detection / Segmentation

`backend/science/vision/segmentation.py` and `backend/science/vision/mask2former.py` provide semantic/panoptic scene parsing, object counts, object coverage, and visualization overlays using a COCO-style 133-class label space. This subsystem serves two roles:

- Direct debug/analysis tool exposed through the Explorer detail modal
- Feature backbone for the affordance models, which depend on object/layout structure rather than raw room labels

Visual Complexity and Fractals

Visual complexity is implemented in `backend/science/math/complexity.py` and computes Shannon entropy, spatial entropy, and edge density. Fractal analysis lives in `backend/science/math/fractals.py` and contributes a fractal-dimension signal consumed by the organized-complexity composite.

Both analyzers are part of the default canonical run path, are persisted to Validation, and are surfaced through the Explorer debug pipeline (edge maps, overlays, and complexity heatmaps). The fractal signal is a scene-level scalar — it is not yet spatially localized as a mask.

Biophilic Materials

The taxonomy workbook does not yet enumerate a dedicated material tag family, but the codebase does. A lightweight biophilia composite is implemented in `backend/science/biophilia.py`, combining plant-presence and natural-texture proxies into `biophilia.index`. In parallel, `backend/science/vision/materials.py` estimates visible materials.

The most clearly biophilic outputs are wood and natural stone families (wood, stone, marble, granite, travertine, terrazzo); brick is also detectable as a natural/mineral surface. The heuristic path uses HSV/luminance/texture cues; the optional Gemini VLM path provides richer named materials, finishes, and location summaries. Outputs are persisted as science attributes and materials_json artifacts, then promoted to canonical material.* tags when estimated coverage is sufficient.

Other Materials

The same materials subsystem supports a broad set of non-biophilic categories: concrete, glass, metal, steel, brass, copper, aluminum, ceramic, tile, porcelain, fabric, leather, vinyl, laminate, plaster, drywall, wallpaper, carpet, paint, plastic, and rubber.

The heuristic path is intentionally cheap and always available, using color, saturation, brightness, local variance, and optional depth weighting to estimate coarse coverage. The optional Gemini path broadens the vocabulary and adds human-readable location and finish metadata. Detections follow the same pipeline path as biophilic materials.

Debug-Mode Localization and Visualization

`backend/api/v1_debug.py` serves edge maps, edge overlays, depth maps, complexity heatmaps, semantic segmentation overlays, panoptic instance overlays, room-result panels, and materials overlays. The Explorer frontend wires these into the detail modal.

Localization sources by output type:

- Objects and affordance cues — segmentation masks, instance labels, counts, and coverage summaries
- Materials — region-specific masks or overlay panels tied to inferred locations
- Complexity — edge maps and heatmaps
- Biophilia — underlying contributing cues are visualizable; the final biophilia.index is a scene-level composite
- Room type — whole-scene classifier; debug overlay shows evidence panels and class distributions, but does not localize the room label to a bounded region

4. Taxonomy Areas Not Currently Supported

Env_Attributes

Env_Attributes is the most ambitious worksheet in the taxonomy, containing 286 rows organized by domain and subdomain. It spans luminous environment variables, thermal factors, acoustics, air quality, ergonomics, temporal qualities, control/agency, topology, and other constructs.

Many rows are not purely visual — some require calibrated photometry, thermal or acoustic sensors, longitudinal occupancy context, or interactive/VR measurement rather than a single RGB frame. Even rows marked as partially monocular often require inverse rendering, depth estimation, or stronger physical priors than the current pipeline provides.

GUIDANCE	Future contributors should treat this sheet as a research roadmap, not a near-term tagging backlog. Promising approaches include: inverse-rendering pipelines for lighting metrics; monocular depth + layout reconstruction for geometric/topological variables; multimodal sensor fusion; VLM-based rubric scoring with explicit uncertainty; and validated proxy models for variables plausibly inferable from still images.

Observable 2D Cues for L01–L58 and All Mediators

L01–L58 of Latent_Variables and all Mediators rows are not yet supported as a coherent implemented layer. These sheets cover broad constructs such as threat, escape efficacy, visibility control, contamination risk, privacy, crowding, restorativeness, fascination, coherence, and control/autonomy.

The Observable_Cues_2D columns are informative, but the cues do not yet collapse into a small set of reusable extractor families in the way that the affordance rows do. Each row would currently require its own mini-research program: identifying viable visual proxies, collecting training/evaluation data, defining a stable output contract, and determining whether the construct is even inferable from a single image.

GUIDANCE	The recommended next step is to impose structure before adding models. Cluster these rows into a small set of cue families (geometry/sightlines, enclosure/partitioning, occupancy density, lighting/surface hygiene, nature/restoration cues), then define one

testable proxy stack per family. From there, a combination of deterministic geometry features, segmentation-derived counts/areas, room priors, and rubric-based VLM scoring could support gradual expansion.

Building_Typologies

Building_Typologies contains 12 high-level program types: Residential, Hospitality, Workplace/Office, Education, Healthcare, Retail, Food & Beverage, Fitness/Wellness, Transportation, Civic/Cultural, Industrial/Back-of-House, and Outdoor/Transitional.

The core difficulty is that building typology is often underdetermined by a single room image. A bedroom suggests Residential or Hospitality; a corridor could belong to a hotel, hospital, office, school, or housing block; a lounge could be residential, workplace, retail, or hospitality. Room type is necessary but frequently insufficient.

GUIDANCE

Future work should likely be hierarchical: infer room type first, then combine with signage/text detection, object priors, material/program cues, and multi-image or album-level context. A scene-graph or retrieval-based approach over sets of images from the same asset collection may be more robust than classifying building typology from a single image in isolation.

5. Exhaustive Current Output Inventory

The table below is exhaustive with respect to the current system contract. For closed tag families (room types, affordances), each supported member is listed explicitly. For open or large namespaces (object.* and material.*), a namespace pattern is used and the backing class lists are enumerated in full after the table.

Tag Group	Tag Name	Confidence?	Magnitude?	Localized?
Room Typology	room_type.bathroom	Yes	No	No
Room Typology	room_type.bedroom	Yes	No	No
Room Typology	room_type.corridor	Yes	No	No
Room Typology	room_type.dining	Yes	No	No
Room Typology	room_type.kitchen	Yes	No	No
Room Typology	room_type.living	Yes	No	No
Room Typology	room_type.office	Yes	No	No
Room Typology	room_type.lobby	Yes	No	No
Room Typology	room_type.retail	Yes	No	No
Room Typology	room_type.classroom	Yes	No	No
Room Typology	room_type.industrial	Yes	No	No
Room Typology	room_type.outdoor_adjacent	Yes	No	No

Room Typology	room_type.other	Yes	No	No
Activity Affordance	L059 Sleep / affordance.l059.(medium|high)	Yes	Yes (1–7 score)	Yes (via cue visualizations)
Activity Affordance	L079 Cook / affordance.l079.(medium|high)	Yes	Yes (1–7 score)	Yes (via cue visualizations)
Activity Affordance	L091 Computer Work / affordance.l091.(medium|high)	Yes	Yes (1–7 score)	Yes (via cue visualizations)
Activity Affordance	L130 Casual Conversation / affordance.l130.(medium|high)	Yes	Yes (1–7 score)	Yes (via cue visualizations)
Activity Affordance	L141 Yoga / Stretching / affordance.l141.(medium|high)	Yes	Yes (1–7 score)	Yes (via cue visualizations)
Objects / Segmentation	object.<COCO-133 class>	Yes	Yes (count & coverage)	Yes
Visual Complexity	complexity.shannon_entropy	No	Yes	Indirectly (heatmap)
Visual Complexity	complexity.spatial_entropy	No	Yes	Indirectly (heatmap)
Visual Complexity	complexity.edge_density	No	Yes	Yes (edge map/overlay)
Fractals	fractal.D	No	Yes	No (scene-level scalar)
Composite Science Index	science.visual_richness	No	Yes	Indirectly
Composite Science Index	science.organized_complexity	No	Yes	Indirectly
Biophilia	biophilia.index	Yes (internal confidence)	Yes	Indirectly (via contributing cues)
Biophilic Materials	material.{wood, stone, marble, granite, travertine, terrazzo, brick}	Yes	Yes (coverage area)	Yes
Other Materials	material.{concrete, glass, metal, steel, brass, copper, aluminum, ceramic, tile, porcelain, fabric, leather, vinyl, laminate, plaster, drywall, wallpaper, carpet, paint, plastic, rubber}	Yes	Yes (coverage area)	Yes

COCO-133 Object / Stuff Class List

All classes used by the segmentation and affordance subsystems:

person, bicycle, car, motorcycle, airplane, bus, train, truck, boat, traffic light, fire hydrant, stop sign, parking meter, bench, bird, cat, dog, horse, sheep, cow, elephant, bear, zebra, giraffe, backpack, umbrella, handbag, tie, suitcase, frisbee, skis, snowboard, sports ball, kite, baseball bat, baseball glove, skateboard, surfboard, tennis racket, bottle, wine glass, cup, fork, knife, spoon, bowl, banana, apple, sandwich, orange, broccoli, carrot, hot dog, pizza, donut, cake, chair, couch, potted plant, bed, dining table, toilet, tv, laptop, mouse, remote, keyboard, cell phone, microwave, oven, toaster, sink, refrigerator, book, clock, vase, scissors, teddy bear, hair drier, toothbrush, banner, blanket, bridge, cardboard, counter, curtain, door-stuff, floor-wood, flower, fruit, gravel, house, light, mirror-stuff, net, pillow, platform, playingfield, railroad, river, road, roof, sand, sea, shelf, snow, stairs, tent, towel, tree, tree-merged, wall-brick, wall-concrete, wall-other, wall-panel, wall-stone, wall-tile, wall-wood, water-other, window-blind, window-other, ceiling-merged, door-stuff-merged, floor-other-merged, food-other-merged, furniture-other-merged, ground-other-merged, mountain-merged, plant-other-merged, sky-other-merged, structural-other-merged, textile-other-merged, vegetation.

Recognized Material Categories

All material categories recognized after normalization:

wood, stone, marble, granite, concrete, brick, glass, metal, steel, brass, copper, aluminum, ceramic, tile, porcelain, fabric, leather, vinyl, laminate, plaster, drywall, wallpaper, carpet, paint, plastic, rubber, terrazzo, travertine.